

ST03 ion-channel counterscreen, third chemistry shipment

Final Report

Study BW17_0167 (LIMS B-ACC-2017-02854)

Client: Selverant Therapeutics Inc

Study Director: Dr B. Rowcroft

Report date: 14 June 2017

1 Summary

40 compounds from the Selverant aminoquinazoline series 03 were tested for Selverant Therapeutics Inc in the hERG automated patch clamp. Testing started on 25 May 2017 and results were reported on 14 June 2017. All runs met plate acceptance criteria. 2 of 40 compounds had an IC₅₀ below 1 μ M and should be treated as a clear liability. Most of the set fell in the 1-10 μ M band, so the margin will depend on free therapeutic exposure.

Table 1 Study summary

Metric	Value
Project	BW17_0167
Client	Selverant Therapeutics Inc
Compounds tested	40
Series / chemotype	Selverant aminoquinazoline series 03 (Aminoquinazoline)
Assays	hERG patch clamp
Turnaround (calendar days)	20
Price	GBP 10,900
hERG IC ₅₀ <1 μ M	2
hERG IC ₅₀ 1-10 μ M	35
hERG IC ₅₀ $\geq$ 10 μ M	3
hERG median IC ₅₀ (μ M)	3.7
E-4031 control IC ₅₀ (nM)	16

2 Objective

To rank the screening set for hERG automated patch clamp liability so that Selverant Therapeutics Inc could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one test lot on 25 May 2017 and logged against B-ACC-2017-02854. Identity and concentration were taken as supplied by the client.

4 Methods

hERG automated patch clamp

HEK293 cells stably expressing hERG (Kv11.1) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells were held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final), with 3-5 minutes per concentration.

- Plate acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current amplitude ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- Positive control E-4031 on every run; IC₅₀ must fall within 5-50 nM
- IC₅₀ from a four-parameter logistic fit of mean tail-current inhibition; $n \geq 2$ cells per compound

5 Results

hERG results

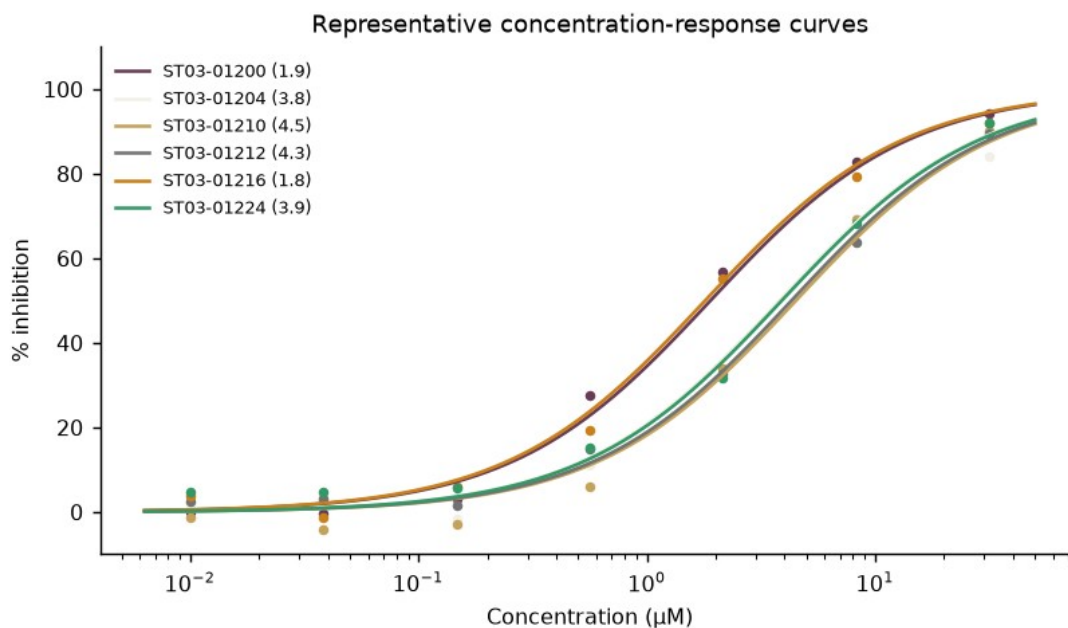
All 40 compounds were tested. 2 of 40 compounds had an IC₅₀ below 1 μ M and should be treated as a clear liability. Most of the set fell in the 1-10 μ M band, so the margin will depend on free therapeutic exposure. The median IC₅₀ was 3.7 μ M (2 below 1 μ M, 35 between 1 and 10 μ M, 3 at or above 10 μ M). E-4031 gave an IC₅₀ of 16 nM, within the acceptance range.

Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ rep 1 (μ M)	IC ₅₀ rep 2 (μ M)	IC ₅₀ rep 3 (μ M)	IC ₅₀ mean (μ M)	% inhibition @10 μ M	n	QC note
ST03-01200	1.9	1.2		1.5	90	2	
ST03-01204	3.8	6.2		4.9	74	2	
ST03-01210	4.5	4.7		4.6	72	2	
ST03-01212	4.3	3.4		3.9	72	2	
ST03-01216	1.8	1.2		1.5	88	2	
ST03-01224	3.9	2.3		3.0	83	2	
ST03-01230	15	7.6		11	46	2	
ST03-01234	4.3	3.8		4.0	64	2	
ST03-01236	6.3	8.0		7.1	55	2	
ST03-01242	2.0	1.1		1.5	80	2	
ST03-01248	2.8	3.5		3.1	74	2	
ST03-01254	4.9	4.0		4.5	70	2	
ST03-01258	8.1	8.3		8.2	59	2	
ST03-01264	1.4	2.4	3.2	2.2	86	3	
ST03-01270	4.9	5.0		4.9	61	2	
ST03-01276	6.6	4.3		5.3	59	2	
ST03-01284	3.2	2.6		2.9	76	2	
ST03-01288	3.5	3.6		3.6	81	2	
ST03-01290	3.5	3.7		3.6	73	2	
ST03-01296	7.5	15		11	50	2	
ST03-01300	6.3	2.2		3.7	79	2	
ST03-01302	4.2	3.6		3.9	72	2	
ST03-01306	4.4	4.0	3.3	3.9	70	3	
ST03-01308	3.3	5.4		4.2	79	2	

ST03-01312	2.8	2.7		2.7	83	2	
ST03-01314	1.1	0.80		0.95	85	2	
ST03-01318	0.57	0.40		0.48	100	2	
ST03-01326	2.1	1.6		1.8	92	2	
ST03-01330	5.2	3.7		4.4	66	2	
ST03-01332	1.7	1.7		1.7	87	2	
ST03-01336	1.7	1.1		1.4	81	2	
ST03-01338	7.4	3.0	4.9	4.8	72	3	
ST03-01342	6.4	3.5		4.7	65	2	
ST03-01344	1.9	3.0	2.4	2.4	80	3	
ST03-01356	1.1	2.3		1.6	85	2	
ST03-01360	3.0	4.2	2.2	3.0	82	3	
ST03-01372	8.9	12		10	50	2	
ST03-01380	4.3	3.1		3.7	79	2	
ST03-01384	2.9	4.0		3.4	76	2	
ST03-01386	4.2	4.7		4.4	70	2	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 16 nM (acceptance 5-50 nM)



6 Discussion

2 of 40 compounds had an IC₅₀ below 1 μ M and should be treated as a clear liability. Most of the set fell in the 1-10 μ M band, so the margin will depend on free therapeutic exposure. These are screening data; binding-site and mechanism statements are outside the scope of this study.

7 Conclusion

The data support ranking within the set. Recommended next step: progress the compounds with the widest hERG margin and no CYP flag, and confirm any CYP3A4 flags by concentration response.

Name	Role	Signature	Date
Bethany Rowcroft	Study Director	<i>Rowcroft</i>	14 June 2017
Ines Alvarado	Assay Quality Coordinator	<i>Alvarado</i>	13 June 2017